

SET- B

1. Write a program to read N integers in an array using pointers and display minimum from them.

```
#include <stdio.h>

int main()
{
    int n,a[20],*ptr,i,*small;
    printf("Enter Size of Array : ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        printf("Enter Number : ");
        scanf("%d",&a[i]);
    }

    small=&a[0];
    for(i=0;i<n;i++)
    {
        if(*(a+i)<*small)
            *small= *(a+i);
    }
    printf("Smallest number in Array is %d",*small);
    return 0;
}
```

Output :

Enter Size of Array : 4

Enter Number : 1

Enter Number : 3

Enter Number : 5

Enter Number : 2

Smallest number in Array is 1

2. Write a function that takes radius of as parameter and two variables. Set first variable to

area of circle and second to perimeter of circle. Accept radius in main and also display area and perimeter in main using the above function.

(Hint: Pass the addresses of the variables to the function to get area and perimeter)

```
#include <stdio.h>

void Calc_Area_and_perimeter(float r,float *a,float *p)
{
    *a=3.142*r*r;
    *p=2*3.142*r;
}

int main()
{
    float r,area,perimeter;
    printf("Enter Radius of circle : ");
    scanf("%f",&r);
    Calc_Area_and_perimeter(r,&area,&perimeter);
    printf("Area of circle is %f",area);
    printf("\nPerimeter of circle is %f",perimeter);
    return 0;
}
```

Output:

Enter Radius of circle : 1.2

Area of circle is 4.524480

Perimeter of circle is 7.540800

3. Write a function which takes distance in kilometer, centimeter and millimeter as parameters as an integer d and increments the distance by d millimeters. Accept distance and d in main and Display the new distance in main using the above function.

```
#include<stdio.h>

void adddistance(float *distance, float *increment, int *flag1);
```

```
int main()
{
    float distance,increment;
    int flag1;
    printf("\n Enter distance :");
    scanf("%f",&distance);
    printf("\n Enter the choice as follows:");
    printf("\n 1 for kilometer\n 2 for centimeter\n 3 for millimeter:");
    scanf("%d",&flag1);
    printf("\n Enter the increment in milimeter:");
    scanf("%f",&increment);
    adddistance(&distance,&increment,&flag1);
    if(flag1==1)
    {
        printf("\n New distance in kilometer is %f",distance);
    }
    else if(flag1==2)
    {
        printf("\n New distance in centimeter is %f",distance);
    }
    else if(flag1==3)
    {
        printf("\n New distance in milimeter is %f",distance);
    }
    else
    {
        printf("\n Invalid choice");
    }
    return 0;
}
```

```

void adddistance(float *distance, float *increment, int *flag1)
{
    if(*flag1==1)
    {
        *distance=((*distance)*1000000);
        *distance=(*distance)+(*increment);
        *distance=(*distance)/1000000;
    }
    else if (*flag1==2)
    {
        *distance=(*distance)*10;
        *distance=(*distance)+(*increment);
        *distance=(*distance)/10;
    }
    else
    {
        *distance=(*distance)+(*increment);
    }
}

```

4. Accept n integers in array A in main. Write a function which takes this array as parameter and find minimum and maximum from it. Display these values in main.

```
#include<stdio.h>
```

```
void find_max_and_min(int a[],int n,int *max,int *min)
```

```
{
```

```
    int i;
```

```
    *max=a[0];
```

```
    *min=a[0];
```

```
    for(i=1;i<n;i++)
```

```
    {
```

```
        if(a[i]< *min)
```

```
        {
```

```
            *min=a[i];
```

```
        }
```

```
    else
```

```
    {
```

```
        *max=a[i];
```

```
    }
```

```
}
```

```
}
```

```
int main()
```

```
{
```

```
    int n,a[20],max,i,min;
```

```
    printf("Enter Size of Array : ");
```

```
    scanf("%d",&n);
```

```
    for(i=0;i<n;i++)
```

```
    {
```

```
        printf("Enter Number : ");
```

```
        scanf("%d",&a[i]);
```

```
}  
find_max_and_min(a,n,&max,&min);  
printf("Max Number is %d and Minimum number is %d",max,min);  
return 0;  
}
```

Output:

Enter Size of Array : 5

Enter Number : 5

Enter Number : 2

Enter Number : 1

Enter Number : 7

Enter Number : 8

Max Number is 8 and Minimum number is 1